Minimum EVM Power Regulation of a Low-Cost 26 GHz SDR mmWave Chain with Sensor-Dominated Loop Dynamics for 5G Systems

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***Abstract*—Millimeter-wave 5G chains must operate at a precise point that preserves signal integrity, yet the attenuator setting that realizes this point drifts with gain and input-power variations. This paper presents a two-stage power-regulation strategy for a low-cost 26 GHz chain built on a sub-6 GHz Software-Defined Radio (SDR) at a 5 GHz intermediate frequency. A 5G NR 256-QAM sweep first maps the Error Vector Magnitude (EVM) offline onto the output of an RMS power detector, reaching a residual EVM of approximately 1.2%; that detector reading is then held in closed loop. A plant model identified from the measured step response shows that the RF chain contributes only a static gain, while the output filtering of the detector sets the entire second-order dynamic, with one pole predicted directly from the detector network without fitting. A deadband proportional-integral regulator restores the reference within the available actuation range; the measured EVM returns below the adopted quality margin.**

***Index Terms*—5G, millimeter-wave (mmWave), Software-Defined Radio, system identification, transfer function, EVM, closed-loop power control, PI controller.**

I. Introduction

The deployment of 5G New Radio (NR) compels the industry toward millimeter-wave (mmWave) bands above 24.25 GHz, offering massive bandwidth and supporting high-order modulation like 256-QAM [1]. However, passive device losses, mixer gain variations [2], and PA non-linear distortion under high-PAPR signals [3] require operating the RF chain at a precise point to preserve signal integrity. In practice, this attenuator setting drifts with temperature, aging, and input-power variations, degrading the Error Vector Magnitude (EVM).

Maintaining this point requires a closed-loop power-control strategy, demanding both an accurate location of the optimal point and a dynamic model of the system. Commercial Software-Defined Radios have made 5G experimentation more accessible [4], but their onboard RF hardware is confined to the sub-6 GHz spectrum and cannot reach millimeter-wave frequencies without additional frequency-translation stages. Recent platforms extend SDR experimentation into the FR2 bands [5], and low-cost setups have been demonstrated for power-amplifier characterization and linearization [6]. These works characterize or linearize at a fixed operating point; none closes the loop to hold it.

Closed-loop power control is well established in production mmWave hardware, where self-heating reduces the effective isotropic radiated power (EIRP) of phased arrays operated at fixed gain [7]. Commercial front-ends address this with an integrated power detector, typically fed by an on-chip directional coupler, whose reading drives a variable-gain stage to hold the output power at a target level [8]. Because the detector is co-integrated with the amplifier, its dynamics are negligible at the loop time scale, and the design reduces to the choice of a loop gain and an error-accumulation weight. In a chain assembled from discrete modules the situation is reversed: the feedback sensor, rather than the RF path, limits the loop. The present work therefore identifies the plant from measured data before designing the regulator, in contrast to [9], where the detector was assumed to contribute no transient response.

Theoretical work in mmWave control often assumes ideal, co-integrated detectors with negligible dynamics. This work instead addresses the theoretical and practical challenges of discrete, low-cost modular chains, where component separation shifts dominant loop dynamics from the RF path to the feedback sensor network. By identifying this sensor-dominated plant and deriving a deadband rule that links continuous

control theory with discrete actuator quantization, it provides a reproducible framework for closing the loop in cost-sensitive deployments.

Its contributions are: (i) a two-stage regulation strategy in which the EVM is mapped offline onto an RMS detector voltage that is then regulated in closed loop, avoiding the real-time demodulation that direct EVM feedback would require; (ii) evidence that in a chain assembled from discrete modules the feedback sensor, not the RF path, sets the plant dynamics, with one pole predicted from the detector output network to within 8% without fitting; and (iii) a deadband rule relating the quantized attenuator step to the calibrated detector slope, which prevents actuator quantization alone from forcing sustained toggling between adjacent attenuation codes.

II. Theoretical Framework

A. Error Vector Magnitude (EVM)

The Error Vector Magnitude (EVM) quantifies the deviation of a measured signal from its ideal constellation point, defining the target operating point. Computed offline during characterization, it is unavailable for real-time control. For N symbols, with measured $S_{meas,n}$ and ideal reference $S_{ref,n}$, EVM is the RMS power ratio of the error vector $\vec{e}_n = S_{meas,n} - S_{ref,n}$ to the reference signal, in percentage:

$$\mathrm{EVM}(\%) = \sqrt{\frac{\frac{1}{N}\sum_{n=1}^{N}|\vec{e}_n|^2}{\frac{1}{N}\sum_{n=1}^{N}|S_{ref,n}|^2}} \times 100\%. \qquad (1)$$

For 256-QAM, 3GPP mandates an EVM below 3.5% [1]; a stricter internal margin of 2.5% is adopted here to preserve signal-quality headroom below the standard limit.

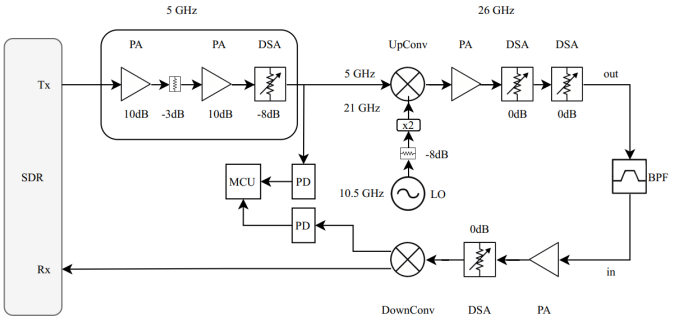
B. Operating-Point Regulation Problem

The output power of an RF chain determines its position in the linearity/noise trade-off: too low a power degrades the signal-to-noise ratio, while too high a power drives the amplifier into compression and raises the EVM. A well-defined minimum-EVM operating point therefore exists, but the attenuator setting that realizes it is not stationary: gain drift due to temperature, bias variation, and component aging displaces the chain from this point over time. Preserving signal quality thus becomes a regulation problem, in which the average output power must be held at a fixed setpoint despite these slow disturbances.

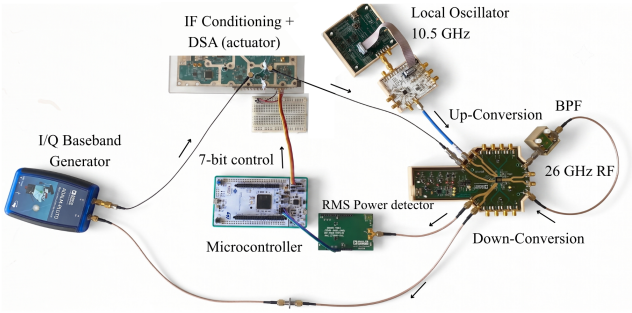
Regulating this quantity requires an observable measurement of the average output power, provided here by an RMS power detector. As Section VI shows, it is the dynamics of that sensor, rather than those of the RF chain, that the controller actually sees.

III. RF Chain Architecture

The setup is a 26 GHz heterodyne RF chain built around an SDR (ADALM-PLUTO) generating and demodulating a 5G NR waveform at a 5 GHz intermediate frequency (IF). The IF signal is conditioned by two 10 dB amplifiers and a 3 dB pad, followed by a digital step attenuator (DSA) setting the



(a) Block diagram of the RF chain.



(b) Laboratory implementation.

Fig. 1: 26 GHz heterodyne RF chain: (a) block diagram and (b) laboratory implementation.

up-conversion mixer drive. A 10.5 GHz local oscillator (LO), internally doubled to 21 GHz, places the signal at 26 GHz. The same LO drives the down-conversion mixer, ensuring common-mode phase noise. The mmWave signal is amplified, passes through two mmWave DSAs and a band-pass filter, and is looped back through an LNA, a receive DSA, and a down-conversion mixer for SDR demodulation. The block diagram and the laboratory implementation are shown in Fig. 1; the nominal attenuator settings used throughout this work are indicated in Fig. 1a.

The actuator of the control loop is the IF-stage DSA (pSemi PE43711), a 7-bit device covering 0 to 31.75 dB, driven here in 0.5 dB steps, which the controller uses to set the power delivered to the up-conversion stage and hence to the millimeter-wave chain. Its 275 ns switching time and 1.6 µs settling time are negligible at the time scale of the loop, as quantified in Section VI.

The feedback signal is provided by an RMS power detector (LTC5596) placed after the down-conversion mixer, which measures the looped-back IF power and therefore observes the millimeter-wave chain through the return path. Unlike a peak detector, it integrates the modulated waveform and outputs a stable DC voltage that varies linearly with the average power expressed in dB, which is required for high-PAPR 5G NR signals without the error introduced by transient envelope peaks. The detector is mounted on an evaluation board whose

output filtering network, as shown in Section VI, sets the dominant dynamics of the control loop.

IV. Methodology

The experimental procedure has two goals: locating the minimum-EVM operating point that defines the control setpoint, and acquiring the transient data used to identify the plant model. Prior to both, the RMS power detector was calibrated by sweeping the SDR output while transmitting a 5 GHz continuous-wave tone and splitting the signal between a spectrum analyzer and the detector, so that the DC voltage read by the microcontroller maps to an absolute power. The response is linear in dB over the operating range, with a measured slope of 28.8 mV/dB ($R^2 > 0.999$), in agreement within 1% with the value specified for the device at this frequency. This calibration makes the detector voltage a validated proxy for absolute power and is used as such throughout.

A. Operating-Point Search

A 5G NR downlink waveform with 20 MHz bandwidth, 30 kHz subcarrier spacing, and 256-QAM modulation was generated using a fixed reference channel configuration (normal cyclic prefix, 24 PRBs in one BWP), transmitted at the 5 GHz IF, looped through the full 26 GHz chain, and demodulated in software to compute the EVM according to (1). For each configuration point, 20 frames were transmitted and a settling interval was allowed between parameter changes. The SDR transmit (Tx) power and receive (Rx) gain were swept over a two-dimensional grid with all attenuators held at the nominal settings of Fig. 1a, so that the sweep isolates the SDR Tx-power/Rx-gain plane, and the resulting EVM was recorded at each point. A coarse sweep first delimited the region of acceptable signal quality, and successive finer sweeps refined the search until the configuration of minimum EVM was isolated. This configuration defines the optimal operating point adopted as the control setpoint.

B. Step-Response Acquisition for Identification

With the chain fixed at the optimal operating point, the plant was excited for identification. A step was applied to the IF-stage DSA, and the resulting transient at the detector output was recorded by the microcontroller as a time series of ADC samples. To characterize repeatability, the experiment was repeated over many runs. The averaged transient provides the data used to fit the plant transfer function in Section VI, while the per-run dispersion of the fitted parameters quantifies the consistency of the identification.

V. Optimal Operating Point

The two-dimensional EVM sweep is shown in Fig. 2. Two regions of severe degradation are visible: one at low Tx power/Rx gain, where the signal collapses into the noise floor, and one at high transmit power, where the chain enters compression and the EVM rises sharply. Between them lies a well-defined operating valley. Successive refinement of the sweep isolated the configuration of minimum EVM at an SDR transmit power of -29.25 dBm and a receive gain of 16 dB, where the chain achieves a residual EVM of about 1.2%, well within the adopted 2.5% margin. This configuration is adopted as the optimal operating point and fixed as the setpoint of the power-control loop.

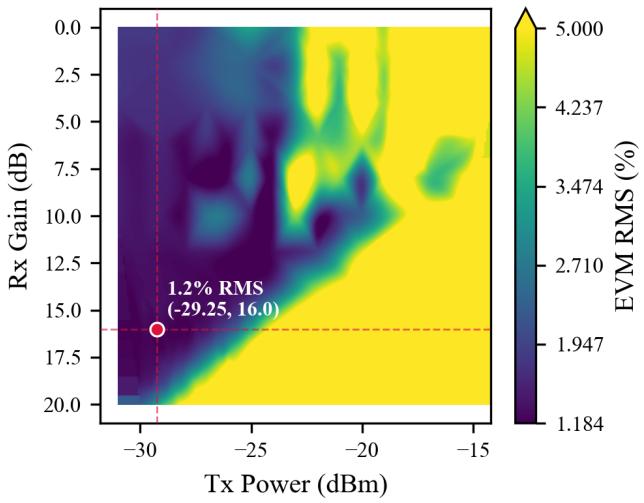


Fig. 2: Two-dimensional EVM map of the 256-QAM waveform. The marker indicates the minimum-EVM point (1.2% RMS at -29.25 dBm SDR Tx power / 16 dB Rx gain) adopted as the control setpoint.

VI. Plant Identification and PI Power Control

Having located the optimal operating point, the chain is now equipped with a closed-loop controller that holds the output power at this setpoint despite disturbances. This section first establishes the control-oriented model of the system, then identifies the plant from its measured step response, designs a proportional-integral (PI) controller on this model, and finally validates the loop experimentally.

A. Control-Oriented System Model

An RF chain conveys information at carrier and symbol rates in the gigahertz range, far beyond the bandwidth of any practical control loop. Closed-loop control therefore acts not on the instantaneous waveform but on the operating point of the chain, that is, its average power. The plant seen by the controller is the cascade of three physically distinct blocks,

$$G(s) = G_{act}(s)\, G_{RF}(s)\, G_{sen}(s), \tag{2}$$

where G_{act}, G_{RF}, and G_{sen} are the actuator, RF-chain, and sensor transfer functions, respectively; G is the overall plant transfer function, and s is the complex frequency variable. Their characteristic times differ by three orders of magnitude.

The actuator is the 7-bit IF-stage digital step attenuator, with a switching time of 275 ns and a settling time of 1.6 μs to within 0.05 dB. At the time scale of the loop it is memoryless: it contributes unity gain in dB/dB, and its relevant non-idealities are quantization and saturation limits (detailed

later in Section VI-C), rather than any dynamic lag. Its glitch-free architecture introduces no transient that could corrupt the identification.

The RF chain maps the commanded attenuation to the power reaching the detector node through the up-conversion, millimeter-wave and down-conversion stages. Operated within its linear region it contributes a static gain $\alpha \approx 1$ in dB/dB and no significant dynamics. Neither the actuator nor the RF chain is separately observable through this sensor; what the measurement establishes is that their combined response lies below the resolution of the detection path, since the recorded transient is accounted for by the sensor network alone.

The sensor is a linear-in-dB RMS detector mounted on an evaluation board whose output network sets the entire dynamic response of the plant. The bare device settles in a few microseconds, whereas the measured transient is two orders of magnitude slower: the dominant pole arises from the 10 nF averaging capacitor at the filter node, and the secondary pole from the 100 Ω/1 μF output network. Two cascaded real poles account for the overdamped second-order behavior identified below. The detector contributes in addition its logarithmic slope m, so that the steady-state gain of the plant is predicted as $K = -\alpha m \approx -28.8$ mV/dB; the negative sign follows from the model and justifies the attenuation-command sign convention in Section VI-C.

B. Plant Identification from the Step Response

To identify the plant, a step variation was applied to the IF-stage digital step attenuator while the resulting transient was recorded at the detector output. The averaged response, shown in Fig. 3, exhibits an overdamped settling behavior that is well captured by a second-order transfer function with two real poles. Relating the detector output to the applied step, the plant is modeled as

$$G(s) = \frac{K}{(\tau_1 s + 1)(\tau_2 s + 1)}, \tag{3}$$

where K is the steady-state gain and τ_1, τ_2 are the dominant and secondary time constants. Fitting (3) to the measured response via a non-linear least squares estimation yields $\tau_1 = 0.273$ ms and $\tau_2 = 0.092$ ms, corresponding to a damping ratio $\zeta = 1.15$ and a natural frequency $\omega_n = 6.32$ rad/ms, and reproduces the experimental transient with a coefficient of determination $R^2 = 0.9999$. As anticipated, this dynamic originates in the detection path rather than in the RF chain. The secondary time constant agrees with the value predicted above from the detector's output network, $RC = 100$ μs, to within 8%, an agreement obtained without fitting that supports the attribution of the plant dynamics to the sensor.

To assess the repeatability of the identification, the step experiment was repeated across many individual runs and the transfer function was extracted from each one. The estimates are tightly clustered around their mean values ($\tau_1 = 0.267$ ms, $\tau_2 = 0.101$ ms, and $\zeta = 1.14$ averaged across runs), confirming that the plant dynamics are consistent across runs and that the model adopted for controller design is statistically representative of the system.

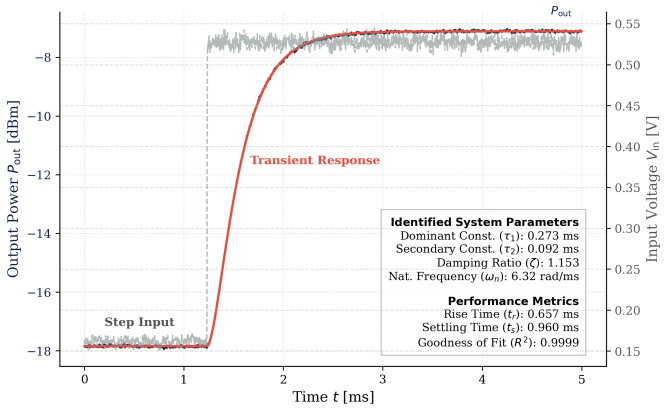


Fig. 3: Measured step response at the detector output (markers) and the fitted second-order transfer function (3) (solid line) used for controller design.

C. PI Controller Design

After identifying the minimum-EVM operating point, the RF chain was equipped with an Automatic Gain Control (AGC)-like power regulation loop. In this scheme, the controller does not use the EVM directly as a real-time feedback variable, since EVM computation requires demodulation and symbol-level post-processing.

Instead, the RMS detector voltage associated with the minimum-EVM operating point is used as the feedback reference. The sweep of Section V therefore determines the optimal operating point, while the control loop maintains the chain around this condition during operation.

The control objective is regulation rather than fast reference tracking. The purpose of the loop is to reject power and gain disturbances that may displace the RF chain from its optimal region.

Such disturbances arise from thermal gain drift, bias variation, and mixer conversion-gain changes. A proportional-integral (PI) controller was adopted, the integral term reducing steady-state error subject to the deadband and available actuation range.

The controller is expressed as

$$C(s) = K_p + \frac{K_i}{s}, \tag{4}$$

where K_p is the proportional gain and K_i is the integral gain. In the digital implementation, the control error is defined as

$$e[k] = V_{\text{ref}} - V_{\text{ADC}}[k], \tag{5}$$

where V_{ref} is the RMS detector voltage corresponding to the optimal operating point and $V_{\text{ADC}}[k]$ is the sampled detector voltage. In this work $V_{\text{ref}} = 0.55$ V, the detector voltage measured at the minimum-EVM condition of Section V.

The controller was deliberately tuned below the bandwidth the identified plant would allow: abrupt attenuation changes act as parasitic amplitude modulation and degrade the EVM of a 256-QAM/OFDM waveform.

To reduce unnecessary control activity around the setpoint, a deadband was introduced in the error path. The effective error used by the PI controller is defined as

$$e_d[k] = \begin{cases} e[k] - \Delta, & e[k] > \Delta, \\ 0, & |e[k]| \le \Delta, \\ e[k] + \Delta, & e[k] < -\Delta, \end{cases} \tag{6}$$

where Δ is the deadband threshold. This strategy sets the effective controller error to zero when the measured detector voltage remains sufficiently close to the reference. Therefore, small fluctuations around the operating point do not immediately generate additional PI corrections, reducing unnecessary switching of the digital step attenuator and rapid variations in the RF output power.

The deadband threshold was selected by considering the interaction between the effective attenuation resolution and the calibrated detector sensitivity. In the implemented controller, the attenuation command is quantized in increments of $q_u = 0.5$ dB. With the calibrated detector slope $m = 28.8$ mV/dB, one effective attenuation increment produces an equivalent steady-state detector-voltage change of

$$q_v = m q_u = 14.4 \text{ mV}. \tag{7}$$

For a reference located between two adjacent attainable attenuation levels, the maximum distance to the nearest attainable steady-state detector voltage is $q_v/2 = 7.2$ mV. Consequently, the selected threshold $\Delta = 8$ mV satisfies

$$\Delta > \frac{q_v}{2}. \tag{8}$$

This condition ensures that at least one attainable quantized operating level can lie inside the deadband. Equivalently, the total deadband width $2\Delta = 16$ mV is slightly greater than the 14.4 mV detector-voltage change associated with one effective DSA increment. Therefore, actuator quantization alone does not necessarily force the controller to alternate between two adjacent attenuation codes.

The digital controller uses an incremental PI form applied directly to the commanded attenuation:

$$u[k] = \text{sat}(u[k-1] - K_p\,(e_d[k] - e_d[k-1]) - K_i T_s\, e_d[k])\,, \tag{9}$$

where $u[k]$ is the attenuation command in dB, T_s is the sampling period, and $\text{sat}(\cdot)$ represents the saturation imposed by the attenuator limits.

The controller runs on an STM32L4R5 microcontroller, which samples the detector through its 12-bit ADC (0.81 mV/LSB) with $T_s = 5$ μs. The implemented controller gains are $K_p = 5$ dB/V and $K_i = 500$ dB/(V·s), with a deadband threshold of $\Delta = 8$ mV ($\sim 10\times$ the 0.81 mV/LSB ADC noise floor). This absorbs sensor noise, satisfies the quantization bound (8), and remains below typical thermal drifts (> 20 mV). The PI gains were initially selected from the identified plant model and then conservatively refined during closed-loop testing to avoid abrupt attenuation changes and sharp EVM transients.

The attenuation command is constrained to the range from 0 to 31.75 dB and quantized in 0.5 dB increments. In addition, the physical DSA command is updated only when the quantized attenuation level changes. Thus, individual ADC fluctuations do not necessarily produce a hardware update. The loop is sampled fast relative to the identified plant dynamics but is tuned for a low closed-loop bandwidth, keeping the chain near the minimum-EVM region without introducing excessive gain fluctuations.

D. Closed-Loop Experimental Results

The closed-loop regulator was evaluated under large and smaller power perturbations. Large steps were applied as stress tests to assess recovery and actuator limits, rather than to reproduce gradual thermal drift. Fig. 4(a) includes a sustained negative power step exceeding the available 7.5 dB compensation range in the up-conversion chain. The controller reaches its actuator limit, leaving a residual error while the step remains applied; recovery follows its removal. A smaller 3 dB perturbation was applied in the up-conversion chain, providing a controlled and repeatable power displacement to characterize recovery within the available range.

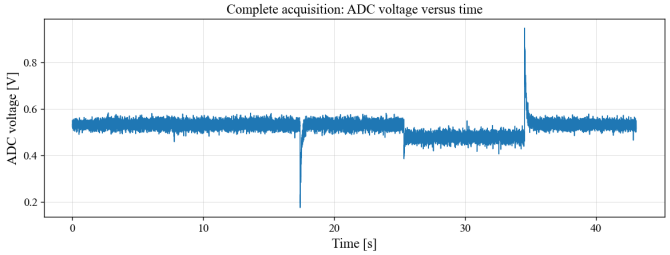
Because the detector is located after down-conversion, the loop cannot distinguish a forward-path gain change from one in the return path. A measurement-path perturbation would instead be compensated in the transmit direction; the reported 3 dB test is a forward-path perturbation.

Fig. 4(b) shows recovery without sustained oscillations, consistent with the conservative tuning adopted to limit abrupt gain changes. Reference recovery is limited by the available actuator range; the residual error in Fig. 4(a) illustrates this limitation.

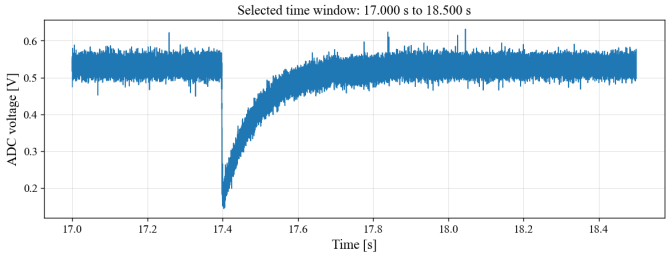
The impact of the closed-loop action on signal quality is evaluated through the RMS EVM computed from the received symbols. The 2.5% margin of Section II is used as the reference limit in the EVM plots.

Fig. 5(a) shows the frame-by-frame RMS EVM during a transient acquisition interval. At the beginning of the acquisition, the RF signal becomes available to the control loop and the controller starts driving the chain toward the selected operating point. During this initial transient, the EVM increases because the chain is temporarily displaced from the minimum-EVM condition. However, as the PI controller restores the detector voltage toward the reference level, the EVM decreases and returns to the low-error region, remaining below the adopted 2.5% margin after stabilization.

A more detailed temporal analysis is shown in Fig. 5(b), where the symbol-level EVM evolution is plotted during the same transient interval. The degradation is concentrated around the initial control action and does not remain permanently after the controller has recovered the setpoint. This behavior indicates that the deadband PI controller drives the RF chain toward the optimal operating region without continuously modulating the RF gain.

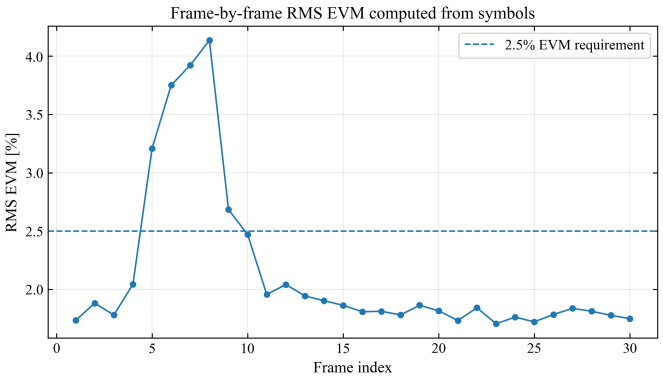


(a) Complete acquisition under large perturbations.

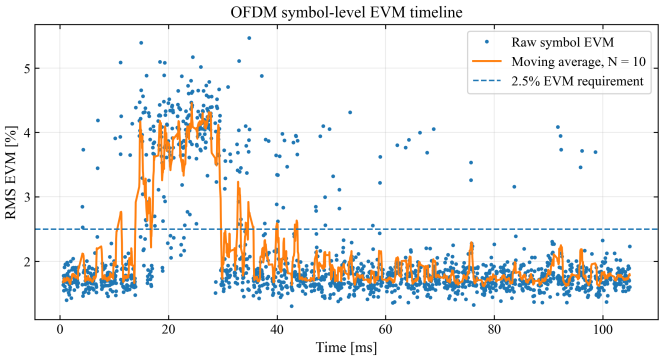


(b) Zoomed response under a 3 dB perturbation.

Fig. 4: Closed-loop RMS detector voltage response: (a) large disturbances, including an actuator-limited interval with residual error; (b) recovery under a 3 dB up-conversion perturbation within the available range.



(a) Frame-by-frame RMS EVM.



(b) Symbol-level EVM timeline.

Fig. 5: EVM behavior during closed-loop regulation. The EVM increases during the initial control transient and subsequently returns to the low-error region.

Overall, the detector reference is recovered within the available actuation range. The displayed post-transient frame-level EVM remains below the adopted margin, supporting regulation around a previously characterized low-EVM condition.

VII. Conclusion

This work presented a minimum-EVM power-regulation strategy for a low-cost 26 GHz RF chain based on a sub-6 GHz SDR operating at a 5 GHz IF with a heterodyne up/down-conversion architecture. The minimum-EVM operating point of the chain was located through a two-dimensional EVM sweep under 5G NR 256-QAM waveforms, at an SDR transmit power of -29.25 dBm with a residual EVM of approximately 1.2%, and was adopted as the control setpoint. From the measured step response, the plant was identified as a second-order system whose dynamics are dominated by the detector's output filtering rather than by the RF chain, with the secondary pole predicted from the sensor network to within 8% without fitting. A deadband PI regulator recovers the detector reference within the available actuation range, with residual error under actuator saturation. The measured post-transient frame-level EVM returns below the adopted quality margin. Future work will extend the loop with disturbance models and additional control strategies, building on preliminary comparisons of classical and intelligent controllers for this chain [10].

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